

Routledge
Taylor & Francis Group

Innovations in Education and Teaching International

ISSN: 1470-3297 (Print) 1470-3300 (Online) Journal homepage: www.tandfonline.com/journals/riie20

University lecturers' perceptions on digital transformation and knowledge management for teaching innovation

Noawanit Songkram, Galih Albarra Shidiq, Winarto, Suphasit Taweejamsup & Nutthakorn Songkram

To cite this article: Noawanit Songkram, Galih Albarra Shidiq, Winarto, Suphasit Taweejamsup & Nutthakorn Songkram (27 Jan 2026): University lecturers' perceptions on digital transformation and knowledge management for teaching innovation, Innovations in Education and Teaching International, DOI: 10.1080/14703297.2026.2621251

To link to this article: https://doi.org/10.1080/14703297.2026.2621251

Published online: 27 Jan 2026.

Submit your article to this journal

Article views: 304

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=riie20

INNOVATIONS IN EDUCATION AND TEACHING INTERNATIONAL
https://doi.org/10.1080/14703297.2026.2621251

Routledge
Taylor & Francis Group

Check for updates

University lecturers' perceptions on digital transformation and knowledge management for teaching innovation

Noawanit Songkram [a,b], Galih Albarra Shidiq [b], Winarto [c], Suphasit Taweejamsup [a,d] and Nutthakorn Songkram [e]

[a]Center of Excellence of Learning Innovation for Thai Society, Chulalongkorn University, Bangkok, Thailand; [b]Department of Educational Technology and Communications, Faculty of Education, Chulalongkorn University, Bangkok, Thailand; [c]Department of Science Education, Faculty of Mathematics and Natural Sciences, Universitas Negeri Yogyakarta, Yogyakarta, Indonesia; [d]Faculty of Law, Chulalongkorn University, Bangkok, Thailand; [e]School of Agricultural Technology, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand

ABSTRACT

This study used the qualitative design to investigate the perceptions of 33 university lecturers through semi-structured interviews and observations conducted after a professional development program in Indonesia. Thematic analysis discovered three interconnected dimensions shaping innovation in teaching: (1) supportive institutional environments that promote responsible behaviour, teamwork, and collaboration; (2) acquisition and application of new knowledge through digital tools and collegial learning; and (3) integration of technological innovation to improve teaching efficiency, creativity, and multidisciplinary engagement. The findings indicate that effective knowledge management enhances lecturers' ability to adapt digital pedagogies, share expertise, and sustain innovation practices. The implications of this study include developing professional learning communities, improving digital literacy programs, and aligning institutional policies to support innovation in teaching. These insights extend global understanding of how universities in emerging contexts can strategically integrate technology and knowledge management to advance educational innovation.

KEYWORDS

Digital transformation; knowledge management; innovation in teaching; digital literacy; higher education

Introduction

The rapid development of digital transformations is reshaping higher education worldwide, requiring universities to rethink how teaching, learning, and knowledge creation are organised. Higher education has captured the challenge: it is not whether technology is used at all, but how universities arrange system-level, data-driven digital transformation that sustains pedagogical innovation and knowledge sharing (Ricardo, 2021). However, in Indonesia, these dynamics have intensified since 2020 through the Merdeka Belajar – Kampus Merdeka (MBKM) reforms and the Eight Key Performance Indicators (IKU) used in performance

CONTACT Galih Albarra Shidiq galihalbarra.s@chula.ac.th Department of Educational Technology and Communications, Faculty of Education, Chulalongkorn University, 254 Phayathai Road, Pathumwan, Bangkok 10330, Thailand

© 2026 Informa UK Limited, trading as Taylor & Francis Group

contracts with universities. These policies have influenced motivations, funding, and reporting mechanisms, compelling institutions to improve digital infrastructure and platforms while increasing professional development that facilitates technology-enhanced pedagogy and knowledge management (Aripradono et al., 2024). Empirically, studies from 2023 to 2025 show universities consolidating online and hybrid practices and tackling organisational capabilities, quality, and equity issues; their pace indicates that digital transformation is ongoing rather than complete (Anggadwita et al., 2024; Rukmini et al., 2023). In this context, innovation in teaching depends on how effectively knowledge is generated, shared, and applied through digital platforms and professional learning environments.

Knowledge management, broadly defined as the systematic process of creating, storing, sharing, and applying knowledge as well as plays an essential role in transforming individual and institutional learning capacities (Nonaka, 1994; Nonaka et al., 1996; Rowley, 2000). Knowledge management and innovation in higher educational environments reflects a critical transformation in pedagogical and research strategies within international academic circumstances (Arpaci, 2017; Fidalgo-Blanco et al., 2015; Veer-Ramjeawon & Rowley, 2020; Zabolotniaia et al., 2020). Within higher education, effective knowledge management practices can improve teaching innovation by connecting lecturers to new pedagogical ideas, supporting digital collaboration, and enabling reflective practice (Fidalgo-Blanco et al., 2015; Veer-Ramjeawon & Rowley, 2020). Similarly, digital transformation defined as strategic integration of digital technologies into teaching and institutional creates opportunities for adaptive learning, data-driven decision-making, and flexible instructional design (Fernández et al., 2023; Mohamed Hashim et al., 2022). However, many developing contexts, including Indonesia, still face challenges such as limited infrastructure, inappropriate digital literacy, and insufficient professional development (Rahman, 2019; Widodo & Riandi, 2013). Furthermore, in Indonesian higher education, technologies offer chances to improve teaching methodologies, learning facilities, management training, and research outcomes through the development of educational content and concepts of knowledge management and innovation for fostering educational experiences (Nur et al., 2017; Pudjiarti et al., 2022; Susanty et al., 2019).

Despite the significant implications of implementing technology into educational practices, university lecturers encounter barriers in teaching and learning digital transformation and knowledge management. These barriers include limited training for professional development that focuses on enhancing their skills and the application of technological resources and ideas (Rahman, 2019; Widodo & Riandi, 2013). Challenges in implementing technologies in Indonesian higher educational institutions are made more difficult by limited digital literacy, particularly in utilising information technology for educational purposes, as well as the quality of teaching methodologies, which includes participation in technology-based training and self-competency development (Santoso et al., 2018; Winoto, 2022). This concern has emerged not only in Indonesia but also within the broader context of global digital information. Fernández et al. (2023) discovered that digital transformation activities in higher education are changing how knowledge is managed, shared, and applied across institutions worldwide. Therefore, effective technology integration requires both infrastructure and supportive institutional contexts that foster innovation and collaboration (Mohamed Hashim et al., 2022).

The invention of technological applications has driven the strategies to promote digital transformation and knowledge management for university lecturers, which is hindered by

the educational institutions' support, which impacts the development of coursework and research in these fields (Leonardi & Treem, 2012; Pudjiarti et al., 2022). Although previous studies have investigated digital technologies and knowledge management in higher education, there is a limited understanding of lecturers' perceptions regarding the integration of digital transformation and knowledge management after structured professional development initiatives in developing higher education systems (Borah & Sidharth, 2022; Donate et al., 2022; González-Mohíno et al., 2024; Lam et al., 2021). However, in Indonesia, these dynamics have particular significance as universities struggle with rapid but inconsistent digital transformation processes and increasing demands to institutionalise knowledge management and promote innovation in pedagogy. This study examines Indonesian university lecturers' views of digital transformation and knowledge management for teaching innovation following their participation in a national professional development program. It contributes through (1) providing an empirically supported overview of lecturers' experiences with institutional environments, learning processes, and technological change, (2) conceptualising knowledge management as a pedagogical framework that integrates digital practices, collegial learning, and institutional support for innovation, and (3) establishing practical implications for the design of professional development and institutional strategies aimed at transforming digital advancements into sustainable teaching innovation.

Literature review

Knowledge management can be defined from various perceptions and has been debated by several researchers. For example, Wetter (2011) defined knowledge management as a process that requires capturing, organising, storing, and transferring knowledge across diverse contexts to foster institutions and economic development through the methodical administration of information, experience, and expertise. In line with this statement, Jansen et al. (2009) argued that transformational leadership styles can facilitate distant leadership, which conveys to institutional members the requirement of enhancing current capabilities within existing domains. Among debates on definitions of knowledge management, Gloet and Terziovski (2004) proposed knowledge management as a formulation to access individual experience, expertise, and skills, which creates new capabilities of innovation, efficacy, and efficiency in a global competitive economy. Similarly, Tian et al. (2009) argued that knowledge management enhances social processes, supports technology resources, and fosters individual behaviours by facilitating and accelerating innovations within educational contexts. A comprehensive analysis of educational research is essential, underlining the significance of knowledge management in enhancing competencies, particularly in pedagogical practices, and facilitating development from the conceptual framework (Darroch, 2005; Du Plessis, 2007; Nonaka, 1994; Rowley, 2000).

Recent studies highlight the critical role of the interplay of knowledge management, intellectual capital, and innovation in enhancing institutional effectiveness within higher education (Ibarra-Cisneros et al., 2023). These findings point out the importance of integrating knowledge management frameworks into academic cultures to enhance innovation and performance. Knowledge management is associated with digital transformation and innovation as well as its impacts on the performance of educational institutions (Lam et al., 2021; Tseng & Lee, 2014). The definition of innovation performance

emphasises the importance of technological and scientific knowledge by promoting activities in key sectors that are linked to innovation output capabilities (Forsman, 2011). The interaction between knowledge management, digital transformation, and innovation is identified by developing individuals or groups' skills and abilities as well as their attitudes and actions (Lam et al., 2021; Mardani et al., 2018; Mehrabani & Shajari, 2012). Researchers have argued that digital transformation and innovation depends on the knowledge conception, which consists of the new ideas designed from deep interaction between individuals and groups in environments to create the knowledge construction, and it has been transformed or implemented in generating values and behaviours (Popadiuk & Choo, 2006).

Methodology

Research design

This study adopted the constructivist theory to enhance understanding of the subject, facilitate paradigm shifts, and provide a completely new viewpoint (Patton, 2014). We employed constructivist theory to examine participants from the university lecturers' perceptions on digital transformation, knowledge management, and teaching innovation within educational institutions, enhancing information and comprehension while addressing evidential, conceptual, and contextual considerations and communication. A qualitative interpretive design grounded in constructivist theory was employed to explore lecturers' lived experiences and perceptions of innovation in teaching. This interpretive model aims to understand research issues by gathering detailed results from one study (Creswell, 2007). This study aims to develop an understanding of the social context by exploring the natural phenomenon that allows university lecturers to reflect on their views, ideas, and experiences concerning various opinions. Constructivism assumes that knowledge is co-constructed through dialogue and reflection (Creswell & Creswell, 2017). This making it well suited for investigating the social and contextual dimensions of pedagogical innovation. Rather than testing hypotheses, this study aimed to develop thick, descriptive insights into how digital transformation and knowledge management interact in educational practice.

Participants and sampling

Thirty-three lecturers from universities across Indonesia participated through purposive sampling. We targeted the participants who (a) had at least 2 year of higher education teaching experience, (b) had participated in the professional development program on knowledge management and digital innovation, and (c) used digital technologies in their teaching or institutional knowledge-sharing activities. We also explain that institutions represented both major urban and regional contexts as presented in Figure 1. Therefore, we justify that these criteria were designed to obtain information-rich cases aligned with the constructivist qualitative design.

Figure 1 illustrates that the participants' groups consisted of 33 university lecturers, with a gender distribution of 30.30% male and 65.70% female. The age distribution was as follows: 18–24 years (30.30%), 25–34 years (51.52%), and 35–44 years (18.18%). A limited number of university lecturers attended the professional development program, with only

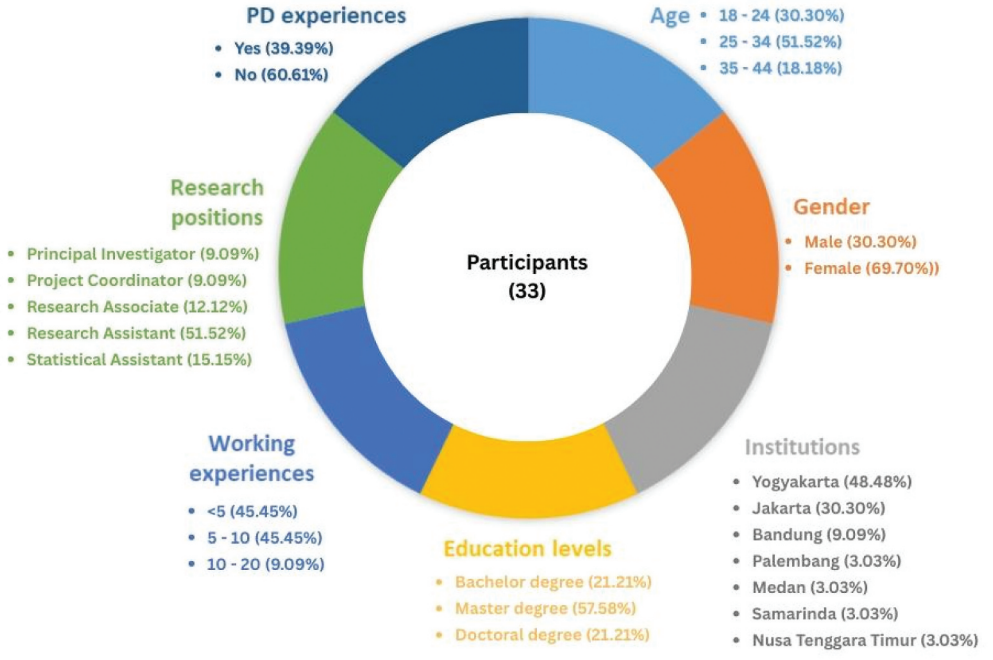


Figure 1. Participants demographic profiles.

39.39% in attendance. Additionally, the university lecturers' educational backgrounds included 57.58% with a master's degree, 21.21% with a doctoral degree, and 21.21% with a bachelor's degree. Furthermore, the university lecturers held diverse research roles: 9.09% served as principal investigators, 9.09% as project coordinators, 12.12% as research associates, 51.52% as research assistants, and 15.15% as statistical assistants. The university lecturers displayed geographical diversity, representing various institutions across Indonesia, with a majority from Yogyakarta (48.48%), followed by Jakarta (30.30%), Bandung (9.09%), Palembang (3.03%), Medan (3.03%), Samarinda (3.03%), and Nusa Tenggara Timur (3.03%). This distribution underlines both regional emphasis and national engagement.

Data collection

This research was conducted in a professional development program for university lecturers from Indonesian universities. The program was established by research institutions and universities in Indonesia. This study employed two qualitative methods: (1) semi-structured interviews extending from 45 to 60 minutes, conducted either online or in person, to investigate lecturers' experiences with knowledge management, digital literacy, and innovation, and (2) non-participant observations during workshops and professional development meetings to record lecturers' interactions, idea sharing, and technology experimentation. We observed group activities and plenary discussions, focussing on participants' interactions, their use of digital tools, and their knowledge-sharing behaviours, which included creating descriptive and analytic field notes and capturing key events, activities, and dialogue among

lecturers. The interview protocol was adapted from Donate and de Pablo (2015), Tseng and Lee (2014), and Gloet and Terziovski (2004) (see Table 1). Interview and observation questions focused on three dimensions: knowledge environment and behaviour, knowledge transfer, application and creation, and innovation performance in teaching and learning.

Data analysis

The purpose of data analysis was to interpret the data to arrive at themes and categories that shed light on how university lecturers perceive the professional development program. In addition to gaining a good general idea of the data, we listened to the tapes of the conversation and read all the transcripts as well as carefully, in a descriptive summary, and discussed the findings with other authors. The data were analysed using qualitative software (NVivo) and interpreted using a thematic analytical method to identify, analyse, and report patterns in the data set of all tapes and transcripts (Creswell & Creswell, 2017; Patton, 2014). The interview transcripts and observation notes were coded within the same NVivo program, allowing observation data to be used for triangulation and for enriching the thematic interpretation of lecturers' narratives. Therefore, we employed a constructivist approach to understanding the distribution of data and developed a coding scheme. The inductive approach was used to analyse the interview, which enhanced the reliability of the analysis. The interview protocol was reviewed, defined and named in an expert discussion. The analysis followed five systematic steps: (1) familiarisation: reading and re-reading all transcripts and field notes to gain an overall understanding of the data, (2) initial coding: generating open codes representing concepts such as knowledge sharing, digital collaboration, innovation in teaching, and institutional support, (3) theme development: grouping related codes into overarching categories aligned with the research questions, e.g. supportive environments, learning new knowledge, technological innovation, (4) review and refinement: themes were reviewed collaboratively among authors to ensure consistency and theoretical alignment with constructivist principles, and (5) interpretation: final themes were synthesised to explain how lecturers perceive knowledge management and digital transformation as enablers of pedagogical innovation.

Results

This study started with an analysis of university lecturers' perceptions of knowledge management, digital transformation, and innovation performance, as expressed through their views, ideas, and experiences during the professional development program. We classified their perspectives into four dimensions. To maintain anonymity, all informants

Table 1. List of questions of the interview.

Dimension	Questions
Knowledge management	What is the purpose of studying knowledge management?
Knowledge environment and behaviour	How you created an environment for responsible behaviour and teamwork to support the research group members?
Knowledge transfer, application, and creation	Explain about each aspect: Are you learning new knowledge? What new knowledge do you receive, and how do you use your ability to apply and improve work efficiency?
Innovation performance	What do you think is an important to use technological innovation in your research? What is the purpose of technological innovation?

are designated by a participant code (e.g. Participant 1, Participant 9, Participant 12). Quotations were chosen to illustrate both common and divergent viewpoints within each theme, representing individuals from various institutional types and areas. Subsequent subsections contain valuable quotations as indented blocks, followed by a concise summary of the speaker's code and relevant context.

Purposes of studying knowledge management

The presented thematic analysis of university lecturers' narrative reflections emphasises the relationship between the purpose of studying knowledge management in the higher educational institutions and the integration of knowledge management into teaching practices and research, as illustrated in Figure 2. These criteria offer critical perceptions on the opportunities and obstacles faced in the examination of knowledge management within educational institutions.

Figure 2 presents the university lecturers' views of the purposes of studying knowledge management. Participants argued that studying knowledge management is related to fostering innovation, improving efficiency, creating new knowledge, developing existing skills, enabling collaboration, enhancing abilities, supporting learning and decision-making, solving problems, promoting sustainability, and addressing gaps in knowledge. The university lecturers' responses suggested that educational institutions should promote and develop the integration of knowledge management, as indicated by the following transcripts.

> Participant 1: I think studying knowledge management could enhance understanding of how knowledge is developed, transferred, organised, and applied within educational and administrative contexts.

> Participant 12: I argue that the key purposes of learning knowledge management are improving efficiency and preventing redundant activities, enhancing learning and teaching, preparing students for the knowledge economy, and facilitating research and innovation through collaboration and community engagement, while establishing decision-making and policy formulation.



Figure 2. Most frequently used words of knowledge management from participants.

> Participant 8: For me, the study of knowledge management is crucial for understanding the phenomena associated with knowledge development

Supportive institutional environments

Creating a comfortable supportive institutional environment is crucial for promoting responsible behaviour and collaboration among university lecturers. The accessibility of resources, tools, and infrastructure in educational institutions significantly affects the willingness of university lecturers to engage in the working culture. Moreover, we should cultivate a culture of accountability, encourage cooperation, teamwork, and leadership as well as ongoing enhancement. Adequate environmental research conditions empower university lecturers optimise their work: teaching efficiency and research activities. Lecturers emphasised that organisational culture and leadership are predominant to determine the capacity for pedagogical innovation are demonstrated in the responses below.

> Participant 4: I feel more motivated to try new teaching strategies with digital tools when my university supports teamwork through regular workshops.

> Participant 7: The working environment is one of the critical aspects of knowledge and behaviour rules that determine how information, ideas, and expertise are shared, including how members interact, collaborate, and maintain ethics. Innovation grows when leadership trusts lecturers to experiment and share what works.

The results of this study demonstrated that providing a knowledge environment and behaviour are fundamental aspects that enable university lecturers to integrate knowledge management and innovation performance in educational environments. These findings provide an example of how the working environment in Indonesian educational institutions is inadequate for facilitating teamwork behaviour. However, the lack of supporting institutional environments could impact university lecturers' abilities and skills, as demonstrated by the following transcripts.

> Participant 9: I think I need more attendance at professional training to improve my capabilities, since I did not receive a supportive working environment.

> Participant 1: The educational institutions in the capital differ from those in local areas regarding their ability to facilitate effective working environments. This might be a barrier hindering university lecturers from promoting research practices and foundational elements, such as setting the stage.

> Participant 5: My personal experience when attending professional development programs, e.g. workplace development, webinars, seminars, and workshops, has enhanced my knowledge to build a network when I meet other university lecturers, and I can also gain more experiences and ideas to share with my team in the workplace.

Learning and applying new knowledge through digital transformation

The results of this study explored the university lecturers' gaining new pedagogical insights in digital collaboration and reflective learning as well as visions of knowledge management, digital transformation, and teaching innovation which were reflected by

their working environment. Knowledge management has been operated as a pedagogical mechanism linking digital infrastructure with creative teaching outcomes.

Figure 3 presents the new knowledge derived from the university lecturers' views and ideas generated from the interviews. According to the arguments, three types of new knowledge have been considered and categorised: knowledge management effectiveness, knowledge management resources, and knowledge members in the new learning that university lecturers can apply and share with their peers. These findings suggestively enhance university lecturers' knowledge management capabilities, specifically in knowledge transfer, application, and creation, which positively impact their innovation performance in educational institutions, as stated in the following transcript.

> Participant 11: I have experience in learning knowledge management effectiveness when using digital tools, which helps me connect with my colleagues from different universities. I learn how they teach and adapt their methods.
>
> Participant 10: I realised I could analyse student learning patterns and adjust my teaching so that I could gain new knowledge while conducting research. At that time, I also shared my ideas to discuss with colleagues. I felt knowledgeable members had a significant impact on my abilities. With more knowledgeable members, I can expand my strategies for working and learning new knowledge, as well as develop my capability in using technologies to assist with my research.
>
> Participant 4: I gained valuable experience when I assumed the role of principal investigator. I have taught knowledge management courses at my university. But attending professional development programs could be a better chance to get collaboration opportunities with other university lecturers.

The results of the interviews indicated that university lecturers actively construct new teaching knowledge through digital exchange rather than passively receiving information. These opportunities are crucial for enhancing my ability to acquire new knowledge and learning. Likewise, the participants also mentioned that knowledge management resources, e.g. organisational structure and culture, along with effective methods of sharing and dissemination, should be regarded as facilitating conditions that enhance

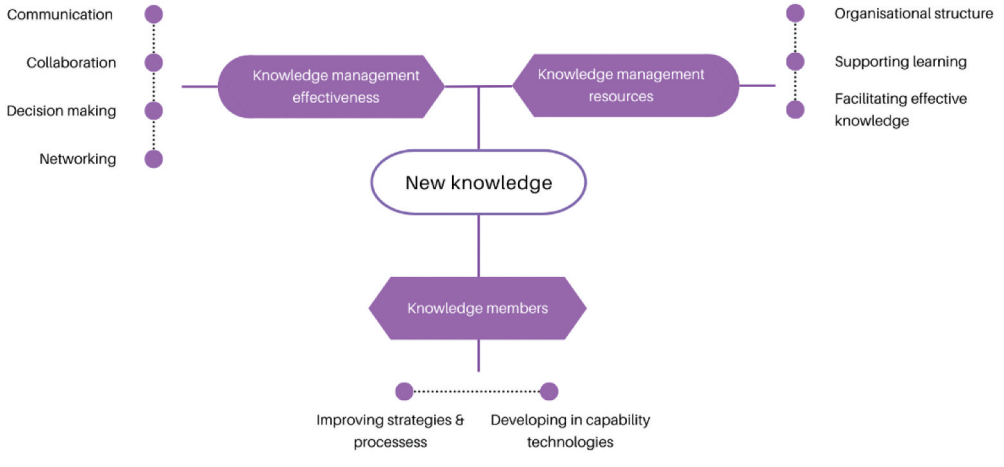


Figure 3. The emerging themes of new knowledge.

the working environment for university lecturers in educational institutions. These responses underline the significant constraints imposed by insufficient knowledge management resources and effectiveness, as well as knowledge members, which should be improved to facilitate university lecturers' environments. The findings of this study highlighted that university lecturers argued that these factors influence their productivity and development skills and abilities through new learning in knowledge management, impacting innovation performance in educational institutions.

Using technological innovation

Technological innovation has integrated in teaching and learning practices, as well as developing research in higher educational institutions. Being innovative is regarded as an opportunity to highlight technological and academic expertise using research and development activities in various sectors associated with the creation of innovation. Figure 4 shows the concepts of technological innovation from university lecturers' views and ideas generated from the interviews.

In this study, we initially analysed university lecturers' awareness of the purposes of technological innovation and the importance of using technological innovation, as well as their impact on the educational institutions. Figure 4 illustrates the various viewpoints of university lecturers, and the transcripts below provide further details.

> Participant 13: I argue that studying technological innovation is crucial to pursuing institutional goals, such as analysing innovation trends, nurturing institutional creativity, and adapting instructional approaches.

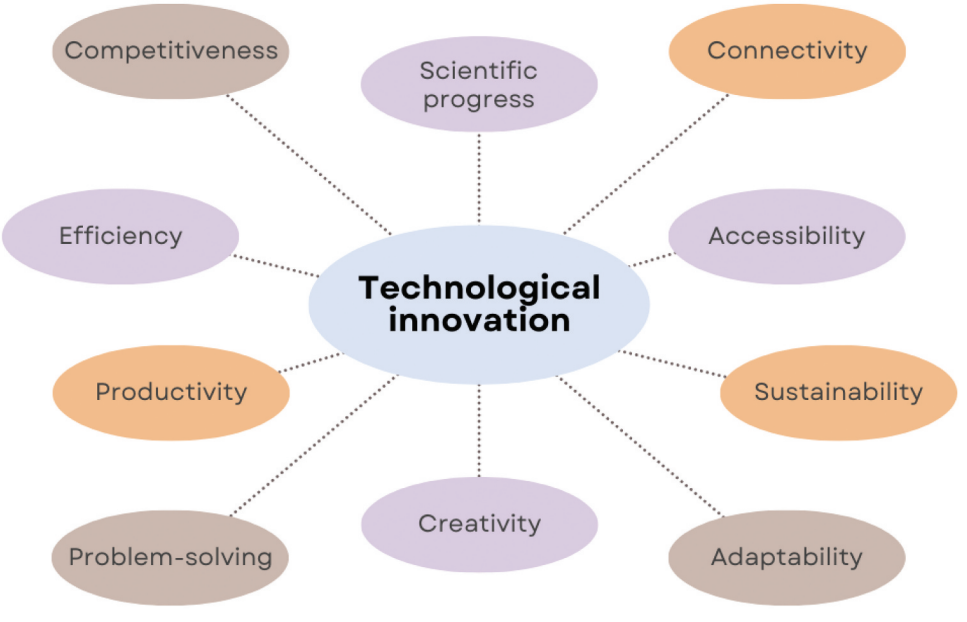


Figure 4. Most used keywords of technological innovation from participants.

> Participant 12: Learning knowledge in technological innovation will enable us to better understand, adapt to, and influence the future of technology in ways that foster growth, sustainability, and social prosperity. For instance, the significance of implementing technology innovation in educational institutions can be examined to enhance learning outcomes, improve management efficiency, foster peer collaboration, support pedagogical approaches, and cultivate a culture of innovation.

> Participant 15: Technological innovation addresses historical development patterns and their effects on human growth, while fostering multidisciplinary learning through the integration of science, engineering, business, and the social sciences.

> Participant 6: Understanding technological innovation is crucial, as it serves as a fundamental catalyst for advancement. It provides the knowledge and insights necessary to analyse how new concepts are created, developed, and integrated into society, ultimately generating value, transforming the environment, and shaping future developments.

The findings of this study indicate that university lecturers reacted favourably to the use of technological innovations in their educational institutions. However, fostering a supportive environment is an important aspect in promoting ongoing innovation, creativity, and collaboration.

Discussion and implications

Supporting a convenient institutional environment could help develop university lecturers' attitudes, abilities, skills, and ideas. The findings of this study explore that innovation in teaching emerges from a dynamic interaction among supportive institutional environments, digital knowledge sharing, and the integration of technological tools. Three major insights contribute to this understanding. The initial finding of this study underlines the importance of creating a supportive institutional environment promotes responsible academic behaviour and teamwork, empowering lecturers to engage collaboratively in designing innovative teaching practices. These results align with previous studies that mentioned the importance of the knowledge environment and behaviour as facilitating knowledge conditions, such as enhancing learning experiences, fostering communication, motivation, and coordination skills, and establishing a supportive team environment for knowledge management concepts (Donate & de Pablo, 2015; Donate et al., 2022; González-Mohíno et al., 2024). The availability of resources, tools, and infrastructure in educational institutions significantly influences the willingness of university lecturers to participate in an innovative working culture. This result supports other studies that have examined similar connections, such as the work of Lam et al. (2021), who found that the creativity of facilitating environmental conditions could create a key understanding of organisational effectiveness, development, and existence. This aligns with Donate and de Pablo (2015), who argue that knowledge-oriented leadership and open communication are essential for transforming shared knowledge into creative instructional approaches. In Indonesian higher education, such environments remain uneven, highlighting the need for policy and resource alignment to sustain collaboration and experimentation in pedagogy.

The second finding of this study explores the acquisition and application of new knowledge through digital tools reflect the constructivist learning cycle, reflection, and application. Lecturers who participated in professional development activities reported

enhanced digital literacy, improved problem-solving, and more confidence in integrating technology into their teaching. These findings align with existing literature, highlighting the significance of knowledge management in developing new knowledge and enhancing work efficiency, which has received attention in the context of knowledge management concepts. For example, Freeze and Kulkarni (2007) explained the idea of knowledge advantage is crucial for assessing the knowledge applied by members, emphasising the various strategies, processes, and technologies employed to efficiently use individual knowledge capabilities. Likewise, Bharadwaj et al. (2015) emphasised the impact of knowledge management is to improve communication, productivity, and skills, e.g. enhancing collaboration and decision-making between individual team members. However, this study also improves knowledge management skills, particularly in knowledge transfer, application, and creation, and positively influences the innovation performance within educational institutions. These findings are supported by the research from Lam et al. (2021), which found a significant relationship between innovation capability and knowledge management, impacting the culture; knowledge outcomes effectively enhance knowledge management approaches for ongoing improvement by using current knowledge.

The third finding of this study discovers that the technological innovation functions not merely as a tool but as a catalyst for pedagogical redesign. When lecturers use digital platforms to support collaborative learning, data-driven feedback, or multimedia instruction, they move from using technology as substitution towards transformation with shifting consistent with the model of technology integration. This study supports the findings of Leonardi and Treem (2012), who argue that using technology effectively can enable strategies to achieve organisational goals, such as improving ongoing organisational practices and promoting effective knowledge management technologies. Kaewsaeng-On et al. (2022) and Pudjiarti et al. (2022) argue that integrating technology into educational environments enhances students' learning experiences and that a creative environment fosters positive behaviour and outcomes. The results of this study point out the interaction of knowledge management practices, technological innovation, and institutional outcomes. The institutional environment that promotes teamwork and responsible behaviour provides a basis for lecturers to collaborate, exchange expertise, and cultivate professional networks. This environment facilitates the acquisition and application of new knowledge, increasingly facilitated by digital tools, collaborative platforms, and training opportunities. Technological innovations enhance these processes by improving efficiency, fostering creativity, and promoting multidisciplinary engagement in teaching and research. Comparable findings have been reported in case studies of higher education institutions globally, indicating that technology adoption and leadership facilitate innovation in teaching and learning (Zhang et al., 2023). From this study, we argue for considering knowledge management as a pedagogical tool that integrates digital practices, collaborative learning, and institutional frameworks for educational innovation.

Conclusion and limitations

This study investigated university lecturers' perceptions of knowledge management and innovation in Indonesian higher education, emphasising the important impact of institutional environments, digital literacy, and accessibility to technological

resources. By integrating digital transformation and knowledge management through a constructivist lens, this study demonstrates how lecturers act as agents of pedagogical innovation. It exposes that sustainable educational transformation depends not only on technology availability but also on cultivating collaborative environments, continuous learning, and institutional cultures that value knowledge sharing. These insights contribute directly to innovative teaching mission by illustrating how universities can translate digital transformation into enduring innovations in teaching and learning.

The study enhances the literature by positioning the relationship between knowledge management and innovation as a critical aspect of digital transformation in higher education. Addressing these interconnected aspects enables universities to enhance knowledge creation and application while integrating with global trends in digitally mediated learning and institutional development. This study extends the discourse on educational innovation by integrating knowledge management theory with digital transformation frameworks in higher education. It conceptualises knowledge management not only as an organisational process but as a pedagogical mechanism that enables continuous learning, collaboration, and innovation among lecturers.

This study shows that the three themes (supporting institutional environments, the acquisition and application of new information, and technological innovation) are inductively determined by the lived experiences of lecturers, rather than immediately corresponding to established models. Also, we emphasise that successful knowledge management and innovation are fundamentally connected to institutional capacity, which promotes digital ability and academic collaboration within the encompassing dialogue on digital transformation in higher education. However, future study could utilise mixed-methods or longitudinal designs to assess the impact of knowledge management-driven digital transformation on student performance and curriculum design. Therefore, comparative analyses across multiple cultural or institutional contexts would enhance theoretical frameworks connecting knowledge management, digital literacy, and pedagogical innovation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research project is supported by Ratchadapiseksompotch Fund and the Second Century Fund (C2F) Chulalongkorn University.

Notes on contributors

Noawanit Songkram is a Professor in the Department of Educational Technology and Communications, Faculty of Education at Chulalongkorn University in Thailand. Her research and professional interests focus on educational and communication technologies, educational innovation development, virtual learning environment, robots for teaching and learning, advanced

statistical analysis, and quantitative research. She has received a Gold Medal and Excellent Awards for educational innovation from several international innovation competitions

Galih Albarra Shidiq is a Postdoctoral Researcher in the Department of Educational Technology and Communications, Faculty of Education at Chulalongkorn University in Thailand and lecturer in Indonesia. His research is focused on educational technology, STEM education, science education, and assessment and evaluation.

Winarto is an Associate Professor in the Department of Science Education, Faculty of Mathematics and Natural Sciences at Universitas Negeri Yogyakarta in Indonesia. His research interests related to science and ethnoscience learning innovation.

Suphasit Taweejamsup is an Associate Professor in Faculty of Law at Chulalongkorn University in Thailand. His research focuses on multiple sectors, i.e. public-private partnerships, state labour policy, aviation management including classroom research. He has strong experience and understanding of knowledge management and interdisciplinary integrated learning.

Nutthakorn Songkram is an Associate Professor of Agricultural Technology and chairperson of the Agricultural Comm§unication Program at King Mongkut's Institute of Technology Ladkrabang. His teaching and research centre on multi-media for learning, electronic media design and development, and information technology.

ORCID

Noawanit Songkram http://orcid.org/0000-0002-7616-2132
Galih Albarra Shidiq http://orcid.org/0000-0002-4889-5419
Winarto http://orcid.org/0000-0001-5029-5327
Suphasit Taweejamsup http://orcid.org/0009-0000-9068-3017
Nutthakorn Songkram http://orcid.org/0000-0003-1197-8218

AI statement

The authors used generative AI for language editing only.

References

Anggadwita, G., Indarti, N., & Ratten, V. (2024). Changes in Indonesian private universities educational practices in the post COVID-19 environment. *The International Journal of Management Education*, *22*(1), 100905. https://doi.org/10.1016/j.ijme.2023.100905

Aripradono, H. W., Nursyamsi, I., Wahab, A., & Sultan, Z. (2024). Educational technology for digital transformation of higher education institutions into entrepreneurial universities. *Policy & Governance Review*, *8*(3), 303–322. https://doi.org/10.30589/pgr.v8i3.1019

Arpaci, I. (2017). Antecedents and consequences of cloud computing adoption in education to achieve knowledge management. *Computers in Human Behavior*, *70*, 382–390. https://doi.org/10.1016/j.chb.2017.01.024

Bharadwaj, S. S., Chauhan, S., & Raman, A. (2015). Impact of knowledge management capabilities on knowledge management effectiveness in Indian organizations. *Vikalpa: The Journal for Decision Makers*, *40*(4), 421–434. https://doi.org/10.1177/0256090915613572

Borah, D., & Sidharth, S. (2022). Analysis of influencing factors on knowledge management in armed forces. In F. H. Pandya, S. Jain, & N. Atchyutuni (Eds.), *Building resilient organizations* (pp. 111–123). Routledge. https://doi.org/10.4324/9781003313663

Creswell, J. W. (2007). *An introduction to mixed-methods research*. University of Nebraska.

Creswell, J. W., & Creswell, J. D. (2017). *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage Publications.

Darroch, J. (2005). Knowledge management, innovation and firm performance. *Journal of Knowledge Management*, *9*(3), 101–115. https://doi.org/10.1108/13673270510602809

Donate, M. J., & de Pablo, J. D. S. (2015). The role of knowledge-oriented leadership in knowledge management practices and innovation. *Journal of Business Research*, *68*(2), 360–370. https://doi.org/10.1016/j.jbusres.2014.06.022

Donate, M. J., González-Mohíno, M., Paolo Appio, F., & Bernhard, F. (2022). Dealing with knowledge hiding to improve innovation capabilities in the hotel industry: The unconventional role of knowledge-oriented leadership. *Journal of Business Research*, *144*, 572–586. https://doi.org/10.1016/j.jbusres.2022.02.001

Du Plessis, M. (2007). The role of knowledge management in innovation. *Journal of Knowledge Management*, *11*(4), 20–29. https://doi.org/10.1108/13673270710762684

Fernández, A., Gómez, B., Binjaku, K., & Meçe, E. K. (2023). Digital transformation initiatives in higher education institutions: A multivocal literature review. *Education and Information Technologies*, *28*(10), 12351–12382. https://doi.org/10.1007/s10639-022-11544-0

Fidalgo-Blanco, Á., Sein-Echaluce, M. L., & García-Peñalvo, F. (2015). Epistemological and ontological spirals: From individual experience in educational innovation to the organisational knowledge in the university sector. *Program*, *49*(3), 266–288. https://doi.org/10.1108/PROG-06-2014-0033

Forsman, H. (2011). Innovation capacity and innovation development in small enterprises. A comparison between the manufacturing and service sectors. *Research Policy*, *40*(5), 739–750. https://doi.org/10.1016/j.respol.2011.02.003

Freeze, R. D., & Kulkarni, U. (2007). Knowledge management capability: Defining knowledge assets. *Journal of Knowledge Management*, *11*(6), 94–109. https://doi.org/10.1108/13673270710832190

Gloet, M., & Terziovski, M. (2004). Exploring the relationship between knowledge management practices and innovation performance. *Journal of Manufacturing Technology Management*, *15*(5), 402–409. https://doi.org/10.1108/17410380410540390

González-Mohíno, M., Donate, M. J., Guadamillas, F., & Cabeza-Ramírez, L. J. (2024). Knowledge-oriented leadership for improved coordination as a solution to relationship conflict: Effects on innovation capabilities. *Knowledge Management Research & Practice*, *22*(4), 388–403. https://doi.org/10.1080/14778238.2024.2306338

Ibarra-Cisneros, M. A., Reyna, J. B. V., & Hernández-Perlines, F. (2023). Interaction between knowledge management, intellectual capital and innovation in higher education institutions. *Education and Information Technologies*, *28*(8), 9685–9708. https://doi.org/10.1007/s10639-022-11563-x

Jansen, J. J. P., Vera, D., & Crossan, M. (2009). Strategic leadership for exploration and exploitation: The moderating role of environmental dynamism. *The Leadership Quarterly*, *20*(1), 5–18. https://doi.org/10.1016/j.leaqua.2008.11.008

Kaewsaeng-On, R., Al-Takhayneh, S. K., Jam, F. A., Chang, B.-L., Pradana, M., & Mahmood, S. (2022). A three wave longitudinal study of school innovation climate and entrepreneurship teachers' acceptance to technology: Moderating role of knowledge sharing and knowledge hiding. *Frontiers in Psychology*, *13*, 1028219. https://doi.org/10.3389/fpsyg.2022.1028219

Lam, L., Nguyen, P., Le, N., & Tran, K. (2021). The relation among organizational culture, knowledge management, and innovation capability: Its implication for open innovation. *Journal of Open Innovation: Technology, Market, and Complexity*, *7*(1), 66. https://doi.org/10.3390/joitmc7010066

Leonardi, P. M., & Treem, J. W. (2012). Knowledge management technology as a stage for strategic self-presentation: Implications for knowledge sharing in organizations. *Information and Organization*, *22*(1), 37–59. https://doi.org/10.1016/j.infoandorg.2011.10.003

Mardani, A., Nikoosokhan, S., Moradi, M., & Doustar, M. (2018). The relationship between knowledge management and innovation performance. *The Journal of High Technology Management Research*, *29*(1), 12–26. https://doi.org/10.1016/j.hitech.2018.04.002

Mehrabani, S. E., & Shajari, M. (2012). Knowledge management and innovation capacity. *Journal of Management Research*, *4*(2), 164–177. https://doi.org/10.5296/jmr.v4i2.1390

Mohamed Hashim, M. A., Tlemsani, I., & Matthews, R. (2022). Higher education strategy in digital transformation. *Education and Information Technologies*, *27*(3), 3171–3195. https://doi.org/10.1007/s10639-021-10739-1

Nonaka, I. (1994). A dynamic theory of organizational knowledge creation. *Organization Science*, *5*(1), 14–37. https://doi.org/10.1287/orsc.5.1.14

Nonaka, L., Takeuchi, H., & Umemoto, K. (1996). A theory of organizational knowledge creation. *International Journal of Technology Management*, *11*(7–8), 833–845.

Nur, R. N. N., Fauzi, A. M., & Sukoco, H. (2017). Strategies of knowledge management implementation for academic services improvement of Indonesian higher education. *Journal of Information & Knowledge Management*, *16*(4), 1750032. https://doi.org/10.1142/S0219649217500320

Patton, M. Q. (2014). *Qualitative research & evaluation methods: Integrating theory and practice*. Sage Publications.

Popadiuk, S., & Choo, C. W. (2006). Innovation and knowledge creation: How are these concepts related? *International Journal of Information Management*, *26*(4), 302–312. https://doi.org/10.1016/j.ijinfomgt.2006.03.011

Pudjiarti, E. S., Lisdiyono, E., & Werdiningsih, R. (2022). Knowledge management to develop comprehensive networking of university-industry collaboration in technology and innovation performance. *International Journal of Data and Network Science*, *6*(2), 461–468. https://doi.org/10.5267/j.ijdns.2021.12.008

Rahman, A. (2019). Restrictive versus facilitative teacher professional development: A case from three public schools in Indonesia. *Asia Pacific Education Review*, *20*(4), 673–686. https://doi.org/10.1007/s12564-019-09585-1

Ricardo, M. A. (2021). Knowledge management and the 2030 agenda for sustainable development in the United Nations context. *Ciencias Administrativas*, *17*(17), 077–84. https://doi.org/10.24215/23143738e077

Rowley, J. (2000). Is higher education ready for knowledge management? *International Journal of Educational Management*, *14*(7), 325–333. https://doi.org/10.1108/09513540010378978

Rukmini, E., Angelina, H., & Anggreni, V. C. (2023). Indonesia higher education's online learning during the pandemic state. *International Journal of Evaluation and Research in Education*, *12*(4), 2286–2301. https://doi.org/10.11591/ijere.v12i4.25103

Santoso, H. B., Lawanto, O., Berliyanto, B., & Janf, P. (2018). Secondary school students' computer self-efficacy and their perception on computer-related tasks in Indonesia. In C. B. Hodges (Ed.), *Self-efficacy in instructional technology contexts* (pp. 245–257). Springer. https://doi.org/10.1007/978-3-319-99858-9_14

Susanty, A. I., Yuningsih, Y., & Anggadwita, G. (2019). Knowledge management practices and innovation performance: A study at Indonesian Government Apparatus Research and Training Center. *Journal of Science and Technology Policy Management*, *10*(2), 301–318. https://doi.org/10.1108/JSTPM-03-2018-0030

Tian, J., Nakamori, Y., & Wierzbicki, A. P. (2009). Knowledge management and knowledge creation in academia: A study based on surveys in a Japanese research university. *Journal of Knowledge Management*, *13*(2), 76–92. https://doi.org/10.1108/13673270910942718

Tseng, S.-M., & Lee, P.-S. (2014). The effect of knowledge management capability and dynamic capability on organizational performance. *Journal of Enterprise Information Management*, *27*(2), 158–179. https://doi.org/10.1108/JEIM-05-2012-0025

Veer-Ramjeawon, P., & Rowley, J. (2020). Embedding knowledge management in higher education institutions (HEIs): A comparison between two countries. *Studies in Higher Education*, *45*(11), 2324–2340. https://doi.org/10.1080/03075079.2019.1608431

Wetter, J. J. (2011). *The impacts of research and development expenditures*. Springer.

Widodo, A., & Riandi. (2013). Dual-mode teacher professional development: Challenges and re-visioning future TPD in Indonesia. *Teacher Development*, *17*(3), 380–392. https://doi.org/10.1080/13664530.2013.813757

Winoto, S. (2022). Improving curriculum and lecturers: Challenges to quality based-technology. *Journal of Social Studies Education Research*, *13*(2), 221–242.

Zabolotniaia, M., Cheng, Z., Dorozhkin, E., & Lyzhin, A. (2020). Retracted article: Use of the LMS Moodle for an effective implementation of an innovative policy in higher educational institutions. *International Journal of Emerging Technologies in Learning (iJET)*, *15*(13), 172–189. https://doi.org/10.3991/ijet.v15i13.14945

Zhang, X., Chen, S., & Wang, X. (2023). How can technology leverage university teaching & learning innovation? A longitudinal case study of diffusion of technology innovation from the knowledge creation perspective. *Education and Information Technologies*, *28*(12), 15543–15569. https://doi.org/10.1007/s10639-023-11780-y